Customer Onboarding – Microservices Backend (Demo)

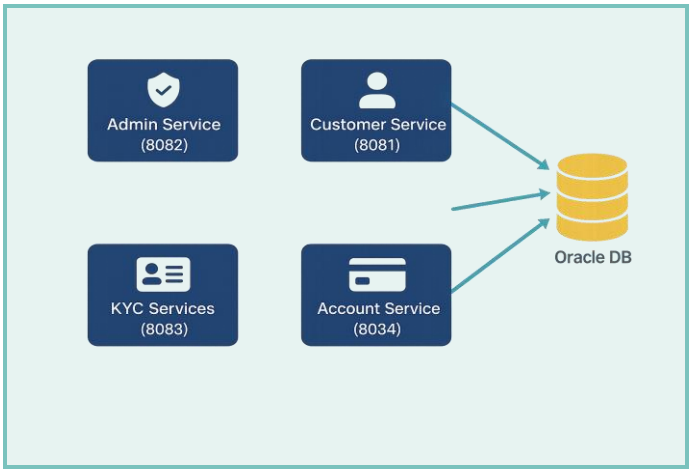
Spring Boot • Oracle • REST • Swagger • Postman

1) Problem & Goal

- Problem: onboarding needs clean data capture, simple KYC, and quick account creation.
- Goal: a minimal microservices backend proving the flow with validations and clear APIs.

2) Architecture Overview

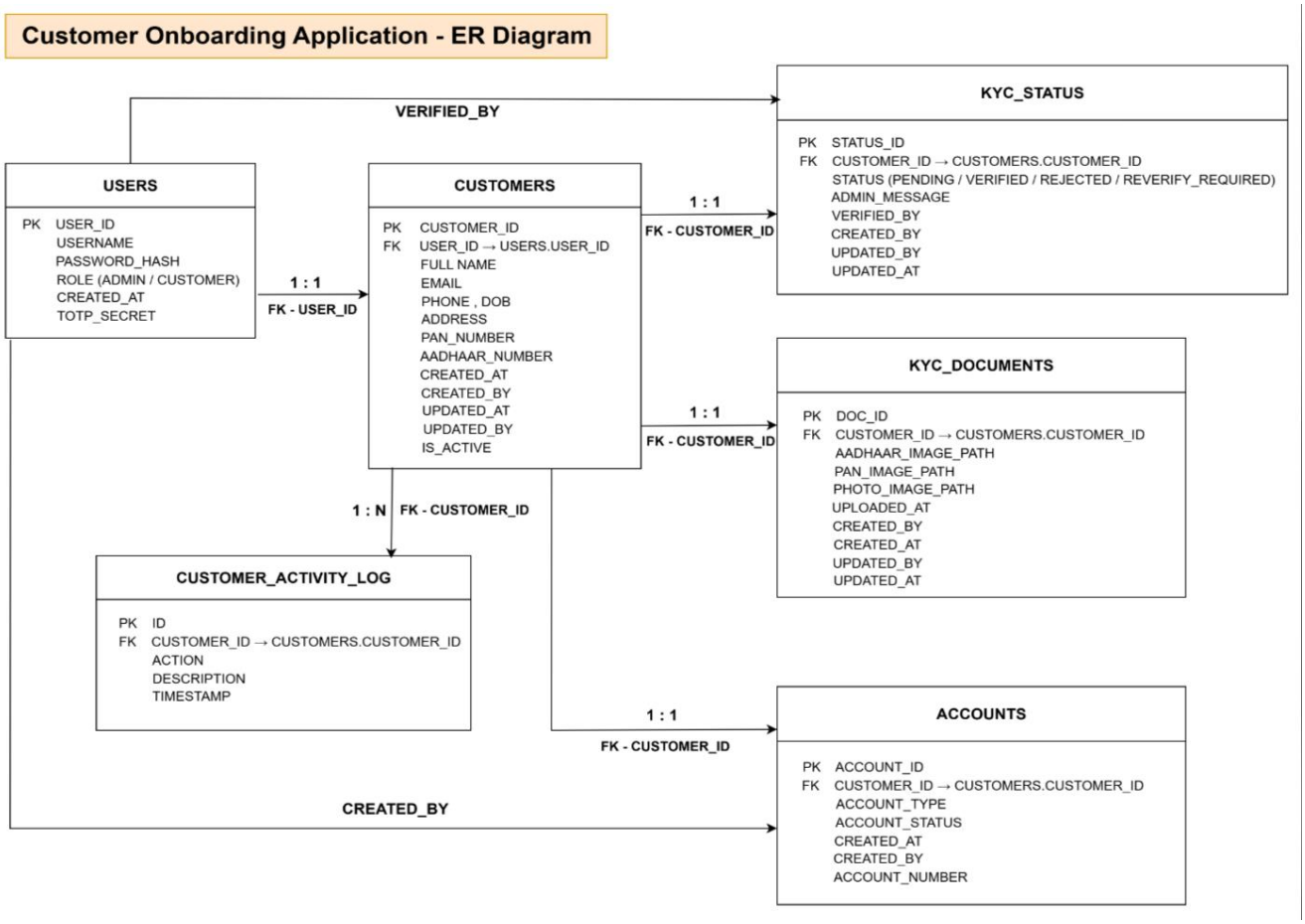
- Admin Service (8082): Authentication and admin KYC review.
- Customer Service (8081): Registration and profile.
- KYC Services (8083): Upload, status, history, downloads.
- Account Service (8084): Account creation, listing, CSV export.
- Oracle DB: persistence for all services (schema: CustomerOnboardingApplication).



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3) Data Model (Key Entities)

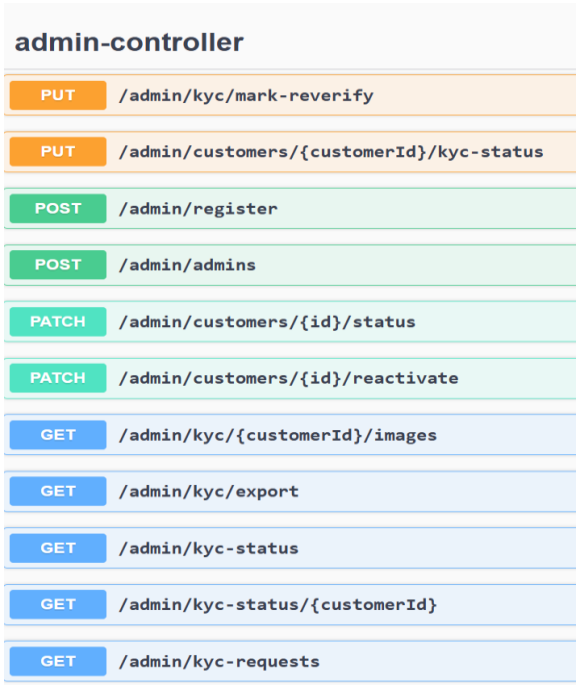
- CUSTOMERS: CUSTOMER_ID, USER_ID, FULL_NAME, EMAIL, PHONE, DOB, PAN_NUMBER, AADHAAR_NUMBER, audit.
- KYC_DOCUMENTS: DOC_ID, CUSTOMER_ID, AADHAAR_IMAGE_PATH, PAN_IMAGE_PATH, PHOTO_IMAGE_PATH, UPLOADED_AT, audit.
- KYC_STATUS: STATUS_ID, CUSTOMER_ID, STATUS, ADMIN_MESSAGE, VERIFIED_BY, VERIFIED_AT, audit.
- ACCOUNTS: ACCOUNT_ID, CUSTOMER_ID, ACCOUNT_TYPE, ACCOUNT_STATUS, ACCOUNT_NUMBER, audit.
- USERS: USER_ID, USERNAME, PASSWORD_HASH, ROLE, TOTP_SECRET, CREATED_AT.

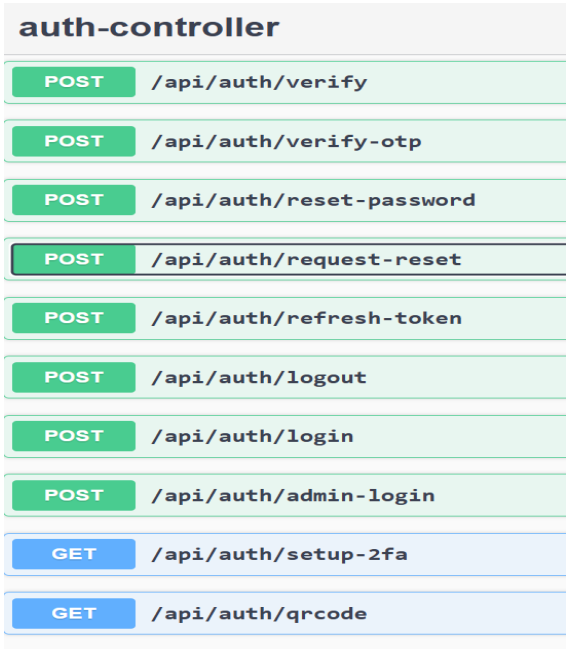


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4) API Surface (Per Service)

- Admin: /api/auth/*, /admin/*
- Customer: /api/customers/*
- KYC: /api/kyc/*
- Account: /api/account/*
- Swagger: /swagger-ui.html on each service





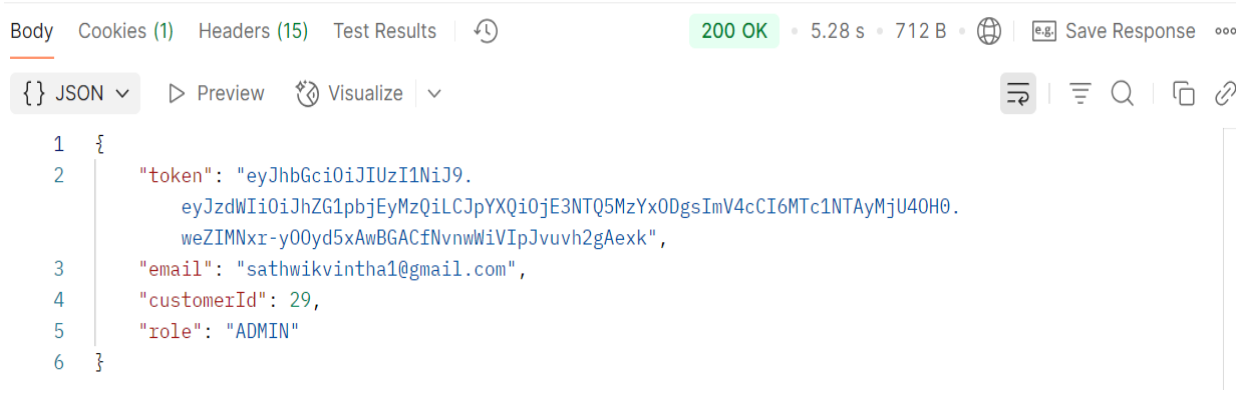
5) Live Demo Plan (Checklist)

- 1) Admin login → JWT token
- 2) Register customer → capture customerId
- 3) Upload KYC (multipart)
- 4) Check KYC status (PENDING)
- 5) (Optional) Mark REVERIFY + Resubmit
- 6) Verify KYC (VERIFIED)
- 7) Create account
- 8) List accounts / Export CSV

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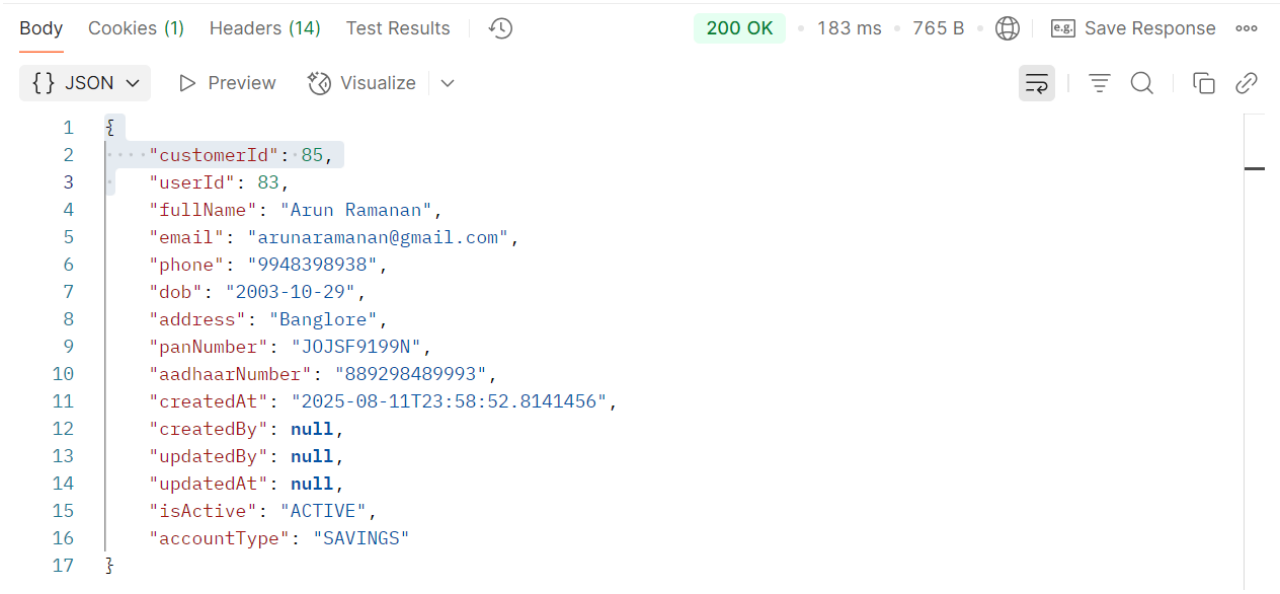
6) Step 1: Admin Login

- POST {{admin_base}}/api/auth/admin-login
- admin_base: http://localhost:8082
- Body: {"username": "admin1234", "password": "admin1234"}
- Expect: 200 with token and role=ADMIN.



7) Step 2: Customer Registration

- POST {{customer_base}}/api/customers/register
- Customer_base: http://localhost:8081
- Validations: username, password, fullname, email, phone(10), PAN, Aadhaar(12), DOB(yyyy-MM-dd), AccountType.
- Expect: 200 with customerId.



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8) Step 3: Upload KYC (Multipart)

- POST {{kyc_base}}/api/kyc/upload
- Form-data: customerId (text), aadhaar (file), pan (file), photo (file).
- Expect: 200 with status=PENDING.

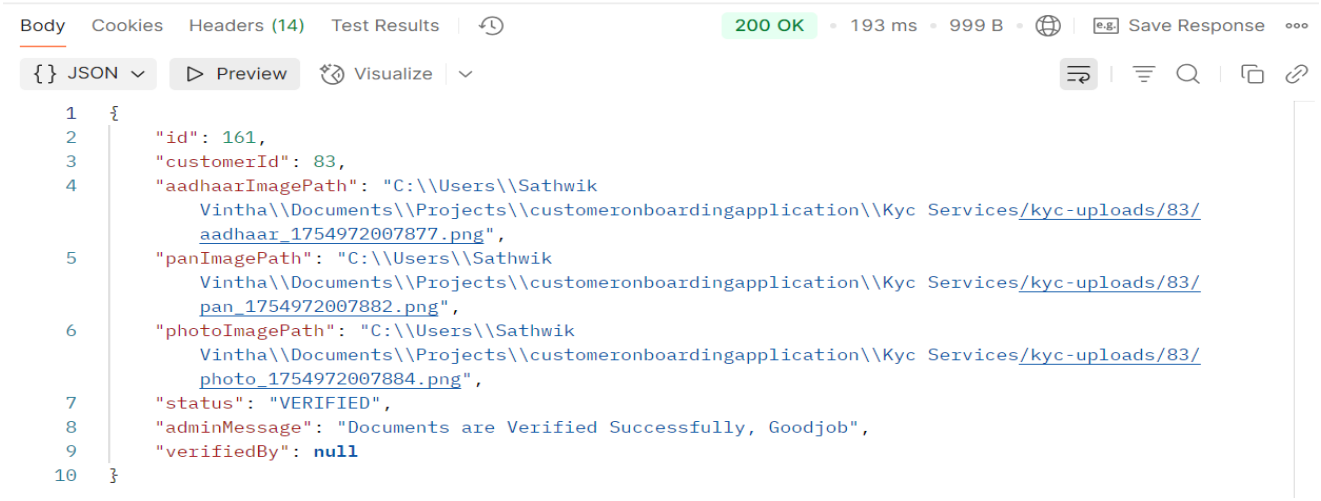


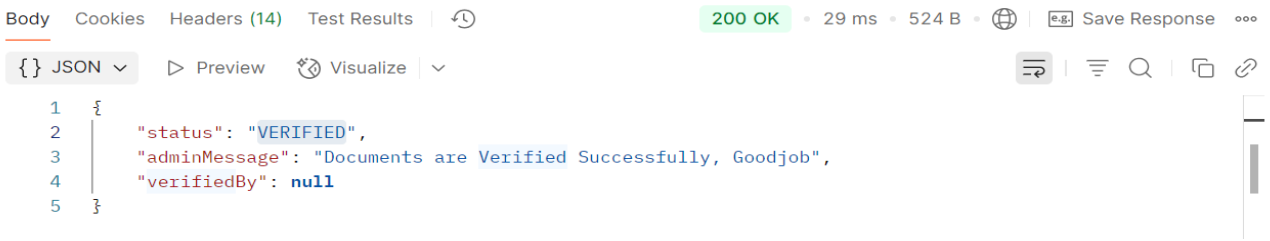
9) Step 4–6: Status → Reverify → Verify

- GET {{kyc_base}}/api/kyc/{{customerId}}/status → PENDING.
- (Optional) PUT {{admin_base}}/admin/kyc/mark-reverify → REVERIFY_REQUIRED.
- PUT {{kyc_base}}/api/kyc/{{customerId}}/status with {"status": "VERIFIED"} → VERIFIED.



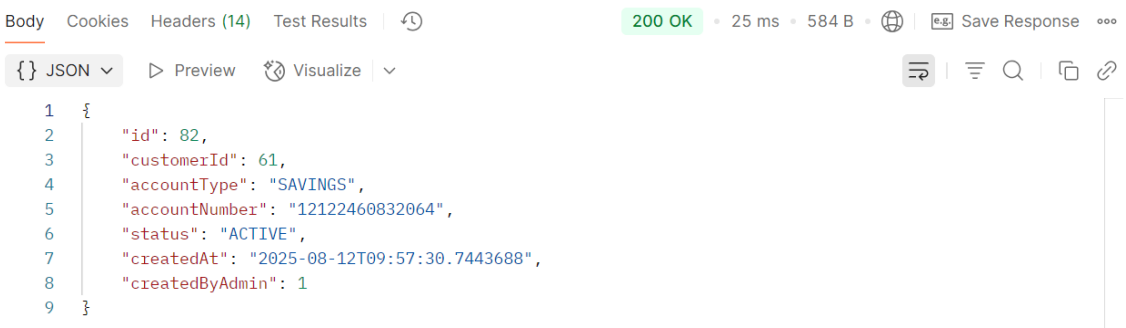
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10) Step 7–8: Create & List Accounts

- POST {{account_base}}/api/account/{{customerId}}?type=SAVINGS&adminId=1 → 200 with accountNumber.
- GET {{account_base}}/api/account/customer/{{customerId}} → list accounts.
- GET {{account_base}}/api/account/export?... → CSV download (optional).



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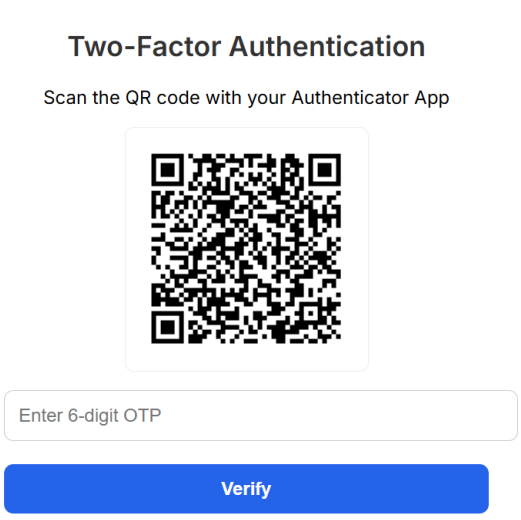


11) Validations & Errors (Show One)

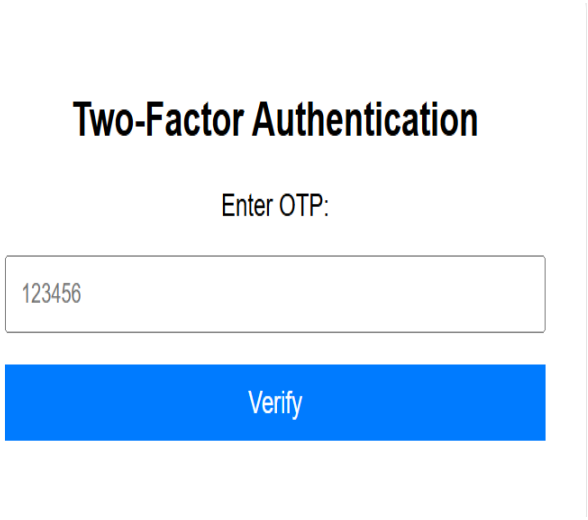
- Invalid PAN/Aadhaar → 400 with field-specific error messages.
- Demonstrates correctness and developer UX.

12) Security & Extensibility

- JWT for protected routes; OTP/TOTP flows in Admin service.



ADMIN LOGIN



CUSTOMER LOGIN

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13) Results & Next

- End-to-end onboarding works: Register → KYC → Verify → Account.
- Next: base64 KYC endpoint, email on REJECT/REVERIFY, automated tests.
- Hardening: DB FKs & indexes; portable file paths.

CustomerOnboardingApplication / Customers (Custom... / **Register Customer (self-service)**

Save | Share

POST http://localhost:8081/api/customers/register **Send**

Params | Authorization | Headers (8) | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Beautify

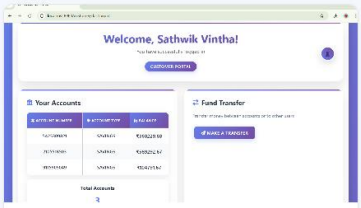
```
{
  "username": "arunaramanan",
  "password": "arunaramanan",
  "fullName": "Arun Ramanan",
  "email": "arunaramanan@gmail.com",
  "phone": "9948398938",
  "dob": "2003-10-29",
  "address": "Banglore",
  "panNumber": "JOJSF9199N",
  "aadhaarNumber": "889298489993",
  "accountType": "SAVINGS"
}
```

KYC Review

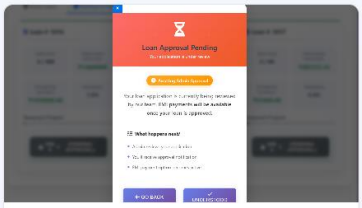
Filter: All

Customer	Full Name	Status	Admin Message	Actions
6	Jack Wills	REVERIFY REQUIRED	Photo Clarity is not Good	Verify, Reject
83	Srinithi Ramesh	VERIFIED	Documents are Verified Successfully, Goodjob	Reverify
1	Sathwik Vintha	VERIFIED	Good, Documents are verified Account has been created Please check it on the Portal	Reverify
61	Nischal Vintha	REJECTED	Image Clarity is not good Please re-upload all the required documents	Verify
2	Prashanth Sai Reddy	PENDING	(null)	Verify, Reject, Reverify

83 Srinithi Ramesh VERIFIED Documents are Verified Successfully, Goodjob Reverify



aadhaar_1754972007877.png Open



pan_1754972007882.png Open

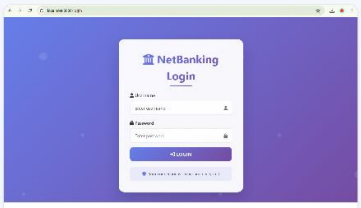
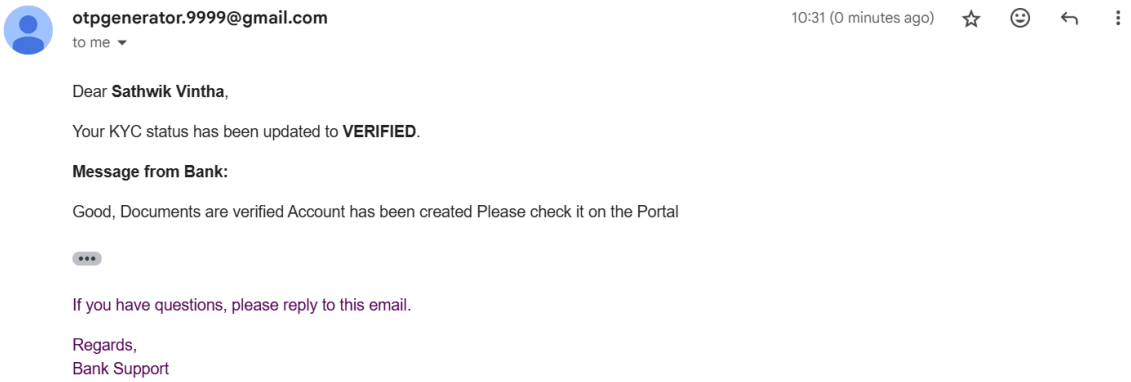


photo_1754972007884.png Open



otpgenerator.9999@gmail.com
to me

10:31 (0 minutes ago)

Dear **Sathwik Vintha**,

Your KYC status has been updated to **VERIFIED**.

Message from Bank:

Good, Documents are verified Account has been created Please check it on the Portal

If you have questions, please reply to this email.

Regards,
Bank Support

Appendix — Screenshot Checklist

- Swagger UI home (Admin).
- Postman: admin login 200 (token).
- Postman: registration 200 (customerId).
- Postman: KYC upload 200 (multipart).
- Postman: KYC status PENDING and VERIFIED.
- Postman: account creation 200 (accountNumber).
- Postman: one 400 validation error (bad PAN).

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